

## Quiz 2:

- |            |                  |                           |
|------------|------------------|---------------------------|
| ◦ textbook | <u>4.5 - 4.8</u> | , End of Chapter Problems |
| ↳ textbook | pg: 132 - 150    | ↳ textbook pg: 167        |
| ↳ pdf      | pg: 159 - 177    | ↳ pdf pg: 194             |
| ↳ ans      | pg: 73 - 76      | ↳ ans pg: 87 -            |
|            |                  | ↳ Q # 4.25 - 4.59         |

# Midterm 2 Prep

## Suggested Problems for Midterm II

Amir Yazdanpour posted on Mar 8, 2026 11:22 AM

Dear all,

The following problems from Chapter 5 are suggested for Midterm II preparation:

5.1-5.5, 5.8, 5.14-5.19, 5.30, and 5.31.

Please note that the last topic of this chapter (Boundary Conditions) will **not** be included in Midterm II and will instead be covered in the next quiz.

Also, the questions from Chapter 4 on the midterm will be similar to those assigned in the bonus assignment.

In addition, some of these problems will be solved during Tuesday's lecture.

## Questions

### • Chapter 5

- # ~~1~~, ~~2~~, ~~3~~, ~~4~~, ~~5~~, ~~8~~ pg 244 (116)
- # ~~14~~, ~~15~~, ~~16~~, ~~17~~, ~~18~~, ~~19~~
- # ~~30~~, ~~31~~

### • Chapter 4

- idk, whatever was on the bonus assignment

## Chapter 5 Questions

5.1) I dens.:  $\mathbf{J} = e^{-x} \cos 4y \hat{\mathbf{a}}_x + e^{-x} \sin 4y \hat{\mathbf{a}}_y$  A/m<sup>2</sup>  
I cross surface? :  $x=2$ ,  $0 < y < \frac{\pi}{3}$ ,  $0 < z < 4$

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{S} \quad , \quad d\mathbf{S} = dx dy dz \\ &= \iiint e^{-x} \cos 4y \hat{\mathbf{a}}_x + e^{-x} \sin 4y \hat{\mathbf{a}}_y \cdot dx dy dz \Big|_{x=2} \\ &= \iint e^{-2} (\cos 4y) dy dz \\ &= e^{-2} \int_0^{\pi/3} \cos(4y) dy \int_0^4 dz \\ &= e^{-2} \cdot \frac{1}{4} \sin(4y) \Big|_0^{\pi/3} \cdot z \Big|_0^4 \\ &= e^{-2} \cdot \frac{1}{4} [\sin(4 \cdot \frac{\pi}{3}) - \sin(4 \cdot 0)] \cdot 4 \\ &= -0.117 \text{ A} \end{aligned}$$

5.2)  $\mathbf{J} = 10 \frac{1}{r} e^{-10^3 t} \hat{\mathbf{a}}_r$  A/m<sup>2</sup>  
I = ? cross surface @  $r=4\text{m}$  @  $t=2\text{ms}$

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{S} \\ &= \iiint 10 \frac{1}{r} e^{-10^3 t} \cdot r^2 \sin \theta \, d\theta \, d\phi \Big|_{t=2\text{ms}, r=4\text{m}} \\ &= 10 \frac{1}{4} e^{-10^3 (2\text{ms})} \cdot (4)^2 \cdot \int_0^{2\pi} \sin \theta \, d\theta \cdot \int_0^\pi d\phi \\ &= 10 \frac{1}{4} e^{-2} \cdot 4^2 \cdot [-\cos \theta]_0^\pi \cdot \phi \Big|_0^{2\pi} \\ &= 40 e^{-2} \cdot 2 \cdot 2\pi \\ &= 68.03 \text{ A} \end{aligned}$$

5.3)  $\mathbf{J} = 10 \frac{1}{\rho} \sin \phi \vec{a}_\rho \text{ A/m}^2$   
 I? thru surface:  $\rho=2, 0 < \phi < \pi, 0 < z < 5$

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{s} \\ &= \iiint 10 \frac{1}{\rho} \sin \phi \cdot \rho \cdot d\rho d\phi dz \quad | \rho=2 \\ &= 10 \int_0^\pi \sin \phi d\phi \int_0^5 dz \\ &= 10 \cdot [-\cos \phi]_0^\pi \cdot z \Big|_0^5 \\ &= 10 \cdot (+2) \cdot (5) \\ &= +100 \text{ A} \end{aligned}$$

5.4) cylindrical conductor  $r=4\text{mm}$ .  $\mathbf{J} = 5e^{-10\rho} \vec{a}_z \text{ A/m}^2$ . I thru conductor?

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{s} \\ &= \iiint 5e^{-10\rho} \rho d\rho d\phi dz \\ &= 5 \int_0^a e^{-10\rho} \rho d\rho \int_0^{2\pi} d\phi \\ &= 5 \cdot \left[ -\frac{1}{10} e^{-10\rho} \cdot \frac{1}{2} \rho^2 \right]_0^a \cdot \phi \Big|_0^{2\pi} \quad , a=4\text{mm} = 4 \times 10^{-3} \\ &= 5 \cdot 7.79 \times 10^{-6} \cdot 2\pi \\ &= 2.45 \times 10^{-4} \text{ A} \end{aligned}$$

5.5)  $\mathbf{J} = 20 \cos \theta \left( \frac{1}{r+3} \right) \vec{a}_r \text{ A/m}^2$

I of surface thru  $r=3, \pi/4 < \theta < \pi/2, 0 < \phi < 2\pi$ ?

$$\begin{aligned} I &= \int \mathbf{J} \cdot d\mathbf{s} \\ &= \iiint 20 \cos \theta \left( \frac{1}{r+3} \right) r^2 \sin \theta dr d\theta d\phi \quad | r=3 \\ &= 20 \frac{1}{r+3} r^2 \cdot \int_{\pi/4}^{\pi/2} \cos \theta \sin \theta d\theta \cdot \int_0^{2\pi} d\phi \\ &= 20 \left( \frac{1}{3+3} \right) 3^2 \cdot \left[ \frac{1}{2} \cos^2 \theta \right]_{\pi/4}^{\pi/2} \cdot \phi \Big|_0^{2\pi} \\ &= 30 \cdot \frac{1}{2} \left( \cos^2 \left( \frac{\pi}{2} \right) - \cos^2 \left( \frac{\pi}{4} \right) \right) \cdot 2\pi \\ &= 30 \cdot \frac{1}{2} (0 - 0.5) \pi \\ &= 47.1 \text{ A} \end{aligned}$$

← ans =  $30\pi$ ? calc error ???

5.8) conducting wire  $r=2\text{mm}$  &  $l=100\text{m}$ .  $V_{DC}=9\text{V}$  applied,  $I=0.3\text{A}$ .

a)  $\mathcal{E}$  field in wire?

b) conductivity of wire?

a)  $\mathcal{E} = \frac{V}{l} = \frac{9}{100} = 0.09 \text{ V/m}$

b)  $R = \frac{l}{\sigma s}$

$\sigma = \frac{l}{R s}$

$= \frac{V/I}{\pi r^2} l$

$= \frac{1/0.3}{\pi (2\text{mm})^2} (100)$

$= 265.258$

$= 265 \text{ k S/m}$

5.14) @ specific temp & pres. He gas has  $5 \times 10^{25}$  atoms/m<sup>3</sup>.  
 10kV/m field applied causes e<sup>-</sup> cloud shift of  $10^{-18}$  m.  
 dielectric constant of He = ?  $\epsilon_r$  = ?

$$|P| = n |p| = n Qd = 2ned = \chi_e \epsilon_0 \epsilon$$

$$2ned = \chi_e \epsilon_0 \epsilon$$

$$\chi_e = \frac{2ned}{\epsilon_0 \epsilon}$$

$$= \frac{2(5 \times 10^{25}) e (10^{-18})}{\epsilon_0 \times 10^4}$$

$$= 1.81 \times 10^{-4}$$

$$* e = \text{const } 23$$

$$\epsilon_0 = \text{const } 32$$

$$\epsilon_r = 1 + \chi_e$$

$$= 1.000181$$

5.15) •  $2 \times 10^{19}$  polar molecules/m<sup>3</sup>  
 • dipole moment:  $1.8 \times 10^{-27}$  C·m  
 • all dipoles aligned in E field direction  
 •  $\vec{E} = 10^5 \vec{a}_x$  V/m  
 •  $P$  = ? ,  $\epsilon_r$  = ?

$$P = \frac{1}{V} \sum_{i=1}^N q_i d_i = \frac{1}{V} \sum_{i=1}^N p_i$$

$$|P| = \frac{1}{V} N |p|$$

$$= 2 \times 10^{19} \cdot 1.8 \times 10^{-27}$$

$$= 3.6 \times 10^{-8}$$

$$\vec{P} = |P| \vec{a}_x$$

$$= 3.6 \times 10^{-8} \vec{a}_x \text{ C/m}^2$$

$$P = \chi_e \epsilon_0 \epsilon E$$

$$\chi_e = \frac{P}{\epsilon_0 \epsilon E}$$

$$= \frac{3.6 \times 10^{-8}}{\epsilon_0 10^5}$$

$$= 0.0407$$

$$\epsilon_r = 1 + \chi_e$$

$$= 1.0407$$

5.16) 10 nC pt charge in wood,  $\epsilon = 4.0$  charge @ origin.  $P$  = ? @  $r = 1$  m

$$\vec{E} = \frac{Q}{4\pi\epsilon_0\epsilon_r r^2} \vec{a}_r$$

$$P = \chi_e \epsilon_0 \epsilon E$$

$$= \frac{\chi_e \epsilon_0 Q}{4\pi\epsilon_0\epsilon_r r^2} \vec{a}_r$$

$$= \frac{3(10 \text{ nC})}{4\pi(4)(1)^2} \vec{a}_r$$

$$= 5.97 \times 10^{-4} \vec{a}_r \text{ C/m}^2$$

5.17) for dielec.  $\epsilon_r = 3.5$ ,  $\vec{P} = 100 \frac{C}{m^2} \hat{a}_P$  nC/m<sup>2</sup>.  
 $\vec{E} = ?$   $\vec{D} = ?$  @  $P = 2m$

$$P = \chi_0 \epsilon_0 \vec{E}$$

$$\vec{\varepsilon} = \underline{p}$$

$$= \frac{100 \cdot \dot{P}}{2.5 \epsilon_0}$$

$$= \frac{100n}{2(2.5)\epsilon_0}$$

$$= 2258.8$$

$$= 2.26 \frac{\text{V}}{\text{Å}}$$

$$D = \epsilon_0 \epsilon_r \epsilon$$

$$= \epsilon_0 (3.5) 2.26 \text{ k} \vec{a}_p$$

$$= 7.00 \times 10^{-8} \text{ } \vec{a}_p \text{ C/m}^2$$

5.18) cylindrical slab w/ polar. w/  $P = p_0 \rho \hat{a}_\rho$ .  
 $p_w = ?$  inside slab & its surface charge density pos?

$$\rho_{pv} = -\nabla p$$

$$= - \frac{1}{p} \frac{\partial}{\partial p} (p p)$$

$$= -\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho \cdot p_0 \cdot \rho)$$

$$= -\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho_0 \rho^2)$$

$$= -\frac{1}{p} (2p \cdot p)$$

$$= -2p_0$$

$$\rho_{p0} = \rho \cdot \hat{a}_p \mid p=a$$

$$= p \cdot p$$

$$= a p.$$

5.19) • spherical shell  $r_i = 1.2 \text{ cm} = 0.012 \text{ m}$  &  $r_o = 2.6 \text{ cm} = 0.026 \text{ m}$ .  
 $\vec{P} = 4 r \vec{a}_r \text{ pC/m}^2$

a) total bound surface charge on inner surface

b)  outer surface

c) " volume charge

a)  $Q_{si} = \int_S P \cdot dS$ ,  $dS = r^2 \sin\theta \, d\theta \, d\phi \, (-\vec{a}_r)$

$$= \iint 4\rho r \cdot r^2 \sin\theta \, d\theta d\phi$$

$$= -4\pi r_i^3 \int \sin\theta \, d\theta \int d\phi$$

$$= -4\rho r_i^3 \int_0^\pi \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$= -4pr_i^3 \cdot -\cos\theta \Big|_0^\pi \cdot \phi \Big|_0^{2\pi}$$

$$= -4\rho r_i^3 \cdot (-\cos(\pi) + \cos(0)) \cdot 2\pi$$

$$= -4(0.012)^3 (2)(2\pi)$$

$$= -8.69 \times 10^{-17} \text{ C}$$

c)  $p_{pv} = -\nabla p$

$$= -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 P_r)$$

$$= -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot 4\pi r)$$

$$= \frac{-1}{r^2} \cdot \frac{\partial}{\partial r} (4pr^3)$$

$$= \frac{-1}{\cancel{5^2}} 12 p \cancel{x^2}$$

$$= -12 p \text{ C/m}^3$$

$$Q_v = \int_v p_{pv} dv$$

$$= \iiint -12\rho \cdot r^2 \sin\theta \, d\theta \, d\phi \, dr$$

$$= -12\rho \int_{0.012}^{0.026} r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi$$

$$= -12 p \cdot \frac{1}{3} r^3 \Big|_{0.012}^{0.026} \cdot -\cos \theta \Big|_0^{\pi} \cdot \phi \Big|_0^{2\pi}$$

$$= -12\rho \cdot \omega \cdot 2 \cdot 2\pi$$

$$= -7.97 \times 10^{-16} \text{ C}$$

$$b) Q_{s2} = -4p r_0^3 \cdot (2)(2\pi)$$

$$= -4\rho (0.026)^3 4\pi$$

$$= -8.83 \times 10^{-16} \text{ C}$$

5.30)  $\vec{J} = e^{-2y} \sin 2x \vec{a}_x + e^{-2y} \cos 2x \vec{a}_y + z \vec{a}_z$  A/m<sup>2</sup>. ROC of electric charge density?

$\nabla \cdot \vec{J} = \frac{\partial}{\partial x} J_x + \frac{\partial}{\partial y} J_y + \frac{\partial}{\partial z} J_z$

Good luck on your midterm!   
-M

$$\begin{aligned} \nabla \cdot \vec{J} &= \frac{\partial}{\partial x} J_x + \frac{\partial}{\partial y} J_y + \frac{\partial}{\partial z} J_z \\ &= \frac{\partial}{\partial x} (e^{-2y} \sin 2x) + \frac{\partial}{\partial y} (e^{-2y} \cos 2x) + \frac{\partial}{\partial z} (z) \\ &= \cancel{e^{-2y} 2 \cdot \cos(2x)} + \cancel{-2e^{-2y} \cos 2x} + 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \frac{\partial \rho_v}{\partial t} &= -\nabla \cdot \vec{J} \\ &= -1 \text{ C/m}^3 \text{ s} \end{aligned}$$

5.31)  $\vec{J} = 100 \frac{1}{\rho^2} \vec{a}_\rho$  A/m<sup>2</sup>

a) time rate of increase in volume charge density

b) the total current passing through surface defined  $\rho = 2$ ,  $0 < z < 1$ ,  $0 < \phi < 2\pi$

$$\begin{aligned} \text{a) } \nabla \cdot \vec{J} &= \frac{1}{\rho} \cdot \frac{\partial}{\partial \rho} (\rho A_\rho) \\ &= \frac{1}{\rho} \cdot \frac{\partial}{\partial \rho} (\cancel{\rho} 100 \frac{1}{\rho^2}) \\ &= \frac{1}{\rho} \cdot \frac{\partial}{\partial \rho} (100 \rho^{-1}) \\ &= \frac{1}{\rho} 100 \cdot -1 \cdot \rho^{-2} \\ &= \frac{1}{\rho} (-100) \cdot \frac{1}{\rho^2} \\ &= -100 \frac{1}{\rho^3} \\ &= -\frac{\partial \rho_v}{\partial t} \\ \frac{\partial \rho_v}{\partial t} &= 100 \frac{1}{\rho^3} \text{ C/m}^3 \text{ s} \end{aligned}$$

$$\begin{aligned} \text{b) } I &= \int \vec{J} \cdot d\vec{S} \\ &= \iint 100 \cdot \frac{1}{\rho^2} \cdot \cancel{\rho} d\phi dz \\ &= 100 \frac{1}{\rho} \int d\phi \int dz \\ &= 100 \frac{1}{\rho} \phi \Big|_0^{2\pi} z \Big|_0^1 \\ &= 100 \frac{1}{2} (2\pi)(1) \\ &= 100\pi \\ &= 314.2 \text{ A} \end{aligned}$$

# Quiz 3 Prep

## Questions:

### • Chapter 6

- # 1, 2, 4, 7, 11
- # 13, 14, 15, 16
- # 29, 30, 31, 66

### • Tutorials

- ?

## Suggested Problems for Final Exam Preparation

Amir Yazdanpour posted on Mar 25, 2026 7:48 PM

Dear all,

The suggested practice problems for the remaining chapters from your textbook are:

5.38–5.42

6.1, 6.2, 6.4, 6.7, 6.11, 6.13–6.16, 6.29–6.31, 6.66

7.1–7.4, 7.6, 7.7, 7.10, 7.16, 7.17, 7.34, 7.37, 7.40, 7.41

8.1–8.9, 8.18–8.21, 8.24–8.26, 8.30–8.34, 8.42–8.46, 8.50, 8.51

These problems are recommended for your final exam preparation. However, the problems that have been solved during lectures, tutorials, midterms, and quizzes are more important and should be your primary focus.

The lectures covering the remaining topics will be completed by Friday, March 27. After that, the remaining lecture time will be dedicated to solving problems from this set and reviewing key concepts.

The final exam will cover Chapters 4–8 (Electrostatics and Magnetostatics). Although you may not see questions directly from Chapters 1–3, as you have noticed, electromagnetics strongly relies on vector calculus, so familiarity with vector operations is still essential.

For those who have limited study time, I recommend prioritizing:

Chapter 6 (Boundary-value problems)

Chapter 7 (Magnetostatics)

Chapter 8 (Magnetic forces and materials)

These chapters are particularly important for the final exam.

## Quiz 3 – Availability and Scope

Amir Yazdanpour posted on Mar 25, 2026 10:27 AM

Dear all,

Quiz 3 will be available this Friday, March 27 at 5:00 PM. The quiz will remain open for 24 hours, but once you start, you will have 45 minutes to complete it.

The content will be from Chapter 6 and will be very similar to the problems solved during the Monday tutorials.

6.1)  $V = 5x^3y^2z$      $\epsilon = 2.25\epsilon_0$      $\vec{E}$  @  $P(-3, 1, 2)$  ?     $\rho_v$  @  $P = ?$

a)  $\vec{E} = -\nabla V$

$$= -\left(\frac{\partial V}{\partial x} \vec{a}_x + \frac{\partial V}{\partial y} \vec{a}_y + \frac{\partial V}{\partial z} \vec{a}_z\right)$$

$$= -\left(\frac{\partial}{\partial x} 5x^3y^2z \vec{a}_x + \frac{\partial}{\partial y} 5x^3y^2z \vec{a}_y + \frac{\partial}{\partial z} 5x^3y^2z \vec{a}_z\right)$$

$$= -\left(15x^2y^2z \vec{a}_x + 2(5)x^3y \vec{a}_y + 5x^3y^2 \vec{a}_z\right)$$

$$\vec{E}_{@P(-3, 1, 2)} = -15(-3)^2(1)^2(2) \vec{a}_x - 10(-3)^3(1)(2) \vec{a}_y - 5(-3)^3(1)^2 \vec{a}_z$$

$$\vec{E} = -270 \vec{a}_x + 540 \vec{a}_y + 135 \vec{a}_z \text{ V/m}$$

b)  $\rho_v = \nabla \cdot D$

$$= \nabla \cdot -\epsilon \nabla V$$

$$= -\epsilon \nabla^2 V$$

$$= -\epsilon \left(\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}\right)$$

$$= -\epsilon \left(\frac{\partial}{\partial x} 15x^2y^2z + \frac{\partial}{\partial y} 10x^3y + \frac{\partial}{\partial z} 5x^3y^2\right)$$

$$= -\epsilon (30xy^2z + 10x^3 + 0)$$

$$\rho_v @ P(-3, 1, 2) = -225\epsilon_0 (30(-3)(1)^2(2) + 10(-3)^3)$$

$$= 1.43 \times 10^{-8} \text{ C/m}^3$$

6.2)  $V = \frac{1}{r^2} 10 \cos \theta \sin \phi$  ,  $\epsilon = \epsilon_0$  . a)  $\vec{E} = ?$  @  $P(1, 60^\circ, 30^\circ)$  . b)  $\rho_v = ?$  @  $P$

6.4) free space.  $V = 10 \rho^{0.8} \text{ V}$ .  $\vec{E} = ?$  &  $\rho_v = ?$  @  $\rho = 0.6 \text{ m}$ .

$$\vec{E} = -\nabla V$$

$$= -\frac{dV}{d\rho} \vec{a}_\rho$$

$$= -\frac{d}{d\rho} 10 \rho^{0.8} \vec{a}_\rho$$

$$= -0.8(10) \rho^{-0.2} \vec{a}_\rho$$

$$= -8 \rho^{-0.2} \vec{a}_\rho$$

$$\vec{E}_{\rho=0.6} = -8(0.6)^{-0.2} \vec{a}_\rho$$

$$= -8.861 \vec{a}_\rho$$

$$\rho_v = -\epsilon_0 \nabla^2 V$$

$$= -\epsilon_0 \frac{1}{\rho} \frac{d}{d\rho} \left( \rho \frac{dV}{d\rho} \right)$$

$$= -\epsilon_0 \frac{1}{\rho} \frac{d}{d\rho} (\rho \cdot 8 \rho^{-0.2})$$

$$= -\epsilon_0 \frac{1}{\rho} \frac{d}{d\rho} (8 \rho^{0.8})$$

$$= -\epsilon_0 \rho^{-1} (0.8)(8) \rho^{-0.2}$$

$$= -6.4 \epsilon_0 \rho^{-1.2}$$

$$\rho_v @ \rho=0.6 = -6.4 \epsilon_0 (0.6)^{-1.2}$$

$$= -1.046 \times 10^{-10} \text{ C/m}^3$$

6.7) in cylinder coords:  $V=0$  @  $\rho=2$  &  $V=60 \text{ V}$  @  $\rho=5$  due to  $\rho_v = \frac{1}{\rho} 10 \rho \text{ C/m}^3$ .  $\epsilon_r = 3.6$ .  $\vec{E} = ?$

$$\nabla^2 V = \frac{-\rho_v}{\epsilon}$$

$$= -\frac{\frac{1}{\rho} 10 \rho}{3.6 \epsilon_0}$$

$$= -\frac{1}{\rho} 0.314 = -\frac{1}{\rho} \frac{d}{d\rho} \left( \rho \frac{dV}{d\rho} \right)$$

$$-0.314 = \frac{d}{d\rho} \left( \rho \frac{dV}{d\rho} \right)$$

$$\int d\rho : \int d\left( \rho \frac{dV}{d\rho} \right)$$

$$\frac{-0.314\rho + A}{\rho} = \frac{dV}{d\rho}$$

$$\frac{dV}{d\rho} = -0.314 + \frac{A}{\rho}$$

$$\int d\rho : \int d\rho$$

$$* V = -0.314\rho + A \ln \rho + B$$

$$60 = -0.314(5) + A \ln(5) + B$$

$$-0 = -0.314(2) + A \ln(2) + B$$

$$60 = -3(0.314) + A \ln\left(\frac{5}{2}\right)$$

$$\frac{60 + 3(0.314)}{\ln\left(\frac{5}{2}\right)} = \frac{A \ln\left(\frac{5}{2}\right)}{\ln\left(\frac{5}{2}\right)}$$

$$A = 66.51$$

$$0 = -2(0.314) + A \ln(2) + B$$

$$B = 2(0.314) - 66.51 \ln 2$$

$$= -45.47$$

$$\vec{E} = -\nabla V$$

$$= -\frac{dV}{d\rho} \vec{a}_\rho$$

$$= -\left(-0.314 + \frac{A}{\rho}\right) \vec{a}_\rho$$

$$\therefore \vec{E} = \left(+0.314 - \frac{66.51}{\rho}\right) \vec{a}_\rho$$

6.11)  $V = x^3 y + y z + c z^2$ .  $c = ?$  satisfying Laplace Eq.

$$0 = \nabla^2 V$$

$$= \frac{\partial^2}{\partial x^2} x^3 y + \frac{\partial^2}{\partial y^2} y z + \frac{\partial^2}{\partial z^2} c z^2$$

$$= \frac{\partial^2}{\partial x^2} 3x^2 y + \frac{\partial^2}{\partial y^2} z + \frac{\partial^2}{\partial z^2} 2c z$$

$$0 = 6xy + 0 + 2c$$

$$2c = -6xy$$

$$c = -3xy$$

6.13) 2 conduct. coaxial cylind. @  $p=1\text{cm}$  &  $p=1.5\text{cm}$ . inner conduct. @ 50V & outer gnd @ 0V.  
separated w/  $\epsilon=4\epsilon_0$ .  $\rho_{s,\text{inner}}=?$

$$0 = \nabla^2 V = \frac{1}{p} \frac{d}{dp} \left( p \frac{dV}{dp} \right)$$

$$V = A \ln p + B$$

$$V(p_2 = 0.015) = 0 = A \ln p_2 + B \rightarrow B = -A \ln p_2$$

$$V(p_1 = 0.010) = 50 = A \ln p_1 + B$$

$$50 = A \ln p_1 - A \ln p_2$$

$$50 = A \ln \left( \frac{p_1}{p_2} \right)$$

$$A = 50 / \ln \left( \frac{p_1}{p_2} \right)$$

$$= -123.3$$

$$V = A \ln p + B$$

$$B = -A \ln p_2$$

$$= -123.3 \ln p + 517.8$$

$$= -123.3 \ln 0.015$$

$$= +517.8$$

$$\vec{E} = -\nabla V$$

$$= -\frac{dV}{dp} \vec{a}_p$$

$$= -\frac{d}{dp} (A \ln p + B) \vec{a}_p$$

$$= -\frac{A}{p} \vec{a}_p$$

$$\rho_s = D_n = \epsilon E_n = \epsilon_r \epsilon_0 \cdot \frac{1}{p} A$$

$$= 4 \epsilon_0 \frac{1}{0.01} (-123.3)$$

$$= 4.37 \times 10^{-7} \text{ C/m}^2$$

6.14) conduct. plates in fig 6.30.

$$V(z=0) = 0 \text{ \& } V(z=2\text{mm}) = 50\text{V} \quad \epsilon_r = 1.5$$

$V=?$   $\vec{E}=?$   $\vec{D}=?$  in region between plates &  $\rho_s$  on plates

$$0 = \nabla^2 V = \frac{d^2 V}{dz^2}$$

$$V = Az + B$$

$$\cdot V(z=0) = 0 : B = 0$$

$$\cdot V(z=d) = V_0 : V_0 = Ad$$

$$A = \frac{V_0}{d}$$

$$V = Az + B$$

$$V = \frac{V_0}{d} z$$

$$= \frac{50}{2\text{mm}} z$$

$$= 25z \text{ KV}$$

$$\vec{E} = -\nabla V$$

$$= -\frac{dV}{dz} \left( \frac{V_0}{d} z \right) \vec{a}_z$$

$$= -\frac{V_0}{d} \vec{a}_z$$

$$= -\frac{50}{2\text{mm}} \vec{a}_z$$

$$= -25 \vec{a}_z \text{ KV/m}$$

$$\vec{D} = \epsilon \vec{E}$$

$$= -\epsilon_r \epsilon_0 \vec{E}$$

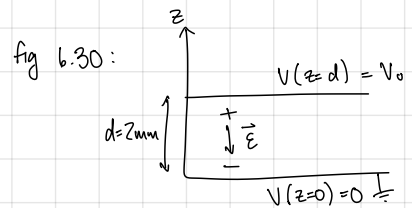
$$= -\epsilon_r \epsilon_0 \frac{V_0}{d} \vec{a}_z$$

$$= -1.5 \epsilon_0 \frac{50}{2\text{mm}} \vec{a}_z$$

$$= -332 \vec{a}_z \text{ nC/m}^2$$

$$\rho_s = D_n$$

$$= \pm 332 \text{ nC/m}^2$$



# Final Exam Prep



## Suggested Problems for Final Exam Preparation ×

Amir Yazdanpour posted on Mar 25, 2026 7:48 PM

Dear all,

The suggested practice problems for the remaining chapters from your textbook are:

5.38–5.42

6.1, 6.2, 6.4, 6.7, 6.11, 6.13–6.16, 6.29–6.31, 6.66

7.1–7.4, 7.6, 7.7, 7.10, 7.16, 7.17, 7.34, 7.37, 7.40, 7.41

8.1–8.9, 8.18–8.21, 8.24–8.26, 8.30–8.34, 8.42–8.46, 8.50, 8.51

These problems are recommended for your **final exam preparation**. However, the problems that have been solved during lectures, tutorials, midterms, and quizzes are more important and should be your primary focus.

The lectures covering the remaining topics will be completed by **Friday, March 27**. After that, the remaining lecture time will be dedicated to solving problems from this set and reviewing key concepts.

The final exam will cover **Chapters 4–8 (Electrostatics and Magnetostatics)**. Although you may not see questions directly from Chapters 1–3, as you have noticed, electromagnetics strongly relies on vector calculus, so familiarity with vector operations is still essential.

For those who have limited study time, I recommend prioritizing:

**Chapter 6 (Boundary-value problems)**

**Chapter 7 (Magnetostatics)**

**Chapter 8 (Magnetic forces and materials)**

These chapters are particularly important for the final exam.

## Questions

### • Chapter 5

• # ~~38~~, ~~39~~, ~~40~~, ~~41~~, ~~42~~

### • Chapter 6

• # ~~1~~, ~~2~~, ~~4~~, ~~7~~, ~~11~~

• # ~~13~~, ~~14~~, ~~15~~, ~~16~~

• # ~~29~~, ~~30~~, ~~31~~, ~~66~~

} from quiz 3 prep

### • Chapter 7

• # ~~1~~, ~~2~~, ~~3~~, ~~4~~, ~~6~~, ~~7~~, ~~8~~

• # ~~10~~, ~~14~~, ~~17~~

• # ~~34~~, ~~37~~, ~~40~~, ~~41~~

← by accident

### • Chapter 8

• # ~~1~~, ~~2~~, ~~3~~, ~~4~~, ~~5~~, ~~6~~, ~~7~~, ~~8~~, ~~9~~

• # ~~18~~, ~~19~~, ~~20~~, ~~21~~, ~~24~~, ~~25~~, ~~26~~

• # ~~30~~, ~~31~~, ~~32~~, ~~33~~, ~~34~~

• # ~~42~~, ~~43~~, ~~44~~, ~~45~~, ~~46~~, ~~50~~, ~~51~~

## Chapter 5

38) region 1:  $z < 0$ ,  $\epsilon_{r1} = 4$   
 region 2:  $z > 0$ ,  $\epsilon_{r2} = 7.5$   
 $\vec{E} = 60\vec{a}_x - 100\vec{a}_y + 40\vec{a}_z$  V/m

a)  $\vec{P}_1 = ?$  b)  $\vec{D}_2 = ?$

a)  $\vec{P}_1 = \chi_{e1} \epsilon_0 \vec{E}_1$

$$= 3 \cdot \epsilon_0 (60, -100, 40)$$

$$= (1.59 \text{ n}, 2.65 \text{ n}, 1.06 \text{ n}) \text{ C/m}^2$$

b)  $\vec{E}_{2t} = \vec{E}_{1t} = 60\vec{a}_x - 100\vec{a}_y$   
 $\vec{D}_{2n} = \vec{D}_{1n} \rightarrow \epsilon_2 \vec{E}_{2n} = \epsilon_1 \vec{E}_{1n}$   
 $\vec{E}_{2n} = \frac{\epsilon_1}{\epsilon_2} \vec{E}_{1n}$   
 $= \frac{4}{7.5} (40\vec{a}_z)$   
 $= 21.3 \vec{a}_z$

$$\vec{E}_2 = 60\vec{a}_x - 100\vec{a}_y + 21.3\vec{a}_z$$

$$\vec{D}_2 = \epsilon_0 \epsilon_{r2} \vec{E}_2 = \epsilon_0 (7.5) (60, -100, 21.3)$$

$$= (3.98 \text{ n}, 6.64 \text{ n}, 1.41 \text{ n}) \text{ C/m}^2$$

39) region 1:  $x > 0$ ,  $\epsilon_1 = 4\epsilon_0$

2:  $x < 0$ ,  $\epsilon_2 = 2\epsilon_0$ ,  $\vec{E}_2 = (6, -10, 8)$  V/m.  $\vec{P}_1 = ?$  &  $\vec{P}_2 = ?$  & densities of 1 & 2 = ?

a)  $\vec{E}_{2t} = \vec{E}_{1t} = (-10, 8)$

$$D_{2n} = D_{1n}$$

$$\epsilon_2 E_{2n} = \epsilon_1 E_{1n}$$

$$E_{1n} = \frac{1}{\epsilon_2} \epsilon_2 E_{2n}$$

$$= \frac{1}{4\epsilon_0} 2\epsilon_0 (\vec{6})$$

$$= 3\vec{a}_x$$

$$E_1 = E_{1t} + E_{1n}$$

$$= (3, -10, 8)$$

$$\vec{P}_1 = \chi_{e1} \epsilon_0 \vec{E}_1$$

$$= 3\epsilon_0 (3, -10, 8)$$

$$= (79.6 \text{ n}, 265.3 \text{ n}, 212.2 \text{ n}) \text{ C/m}^2$$

$$\vec{P}_2 = \chi_{e2} \epsilon_0 \vec{E}_2$$

$$= 1 \cdot \epsilon_0 (6, -10, 8)$$

$$= (53.1 \text{ n}, -88.4 \text{ n}, 70.7 \text{ n}) \text{ C/m}^2$$

b)  $w_1 = \frac{1}{2} \epsilon_1 E_1^2$

$$= \frac{1}{2} (4\epsilon_0) (3^2 + 100^2 + 8^2)$$

$$= 346 \epsilon_0$$

$$= 3.06 \text{ nJ/m}^2$$

$$w_2 = \frac{1}{2} \epsilon_2 E_2^2$$

$$= \frac{1}{2} (2\epsilon_0) (6^2 + 100^2 + 8^2)$$

$$= 200 \epsilon_0$$

$$= 1.77 \text{ nJ/m}^2$$

40)  $4x + 3y = 10$  m

$$\vec{D}_1 = (2, -4, 6.5) \text{ nC/m}^2$$

$$\epsilon_{r2} = 2.5$$

$$\vec{D}_2 = ? \quad \& \quad \theta_2 = ? \quad \vec{D}_2 \text{ makes w/ normal}$$

$$f(x, y) = 4x + 3y - 10 = 0$$

$$\nabla f = 4\vec{a}_x + 3\vec{a}_y$$

$$\vec{a}_n = \frac{-\nabla f}{|\nabla f|} = \frac{-(4, 3)}{\sqrt{4^2 + 3^2}} = (-0.8, -0.6)$$

↳ minus sign cause towards origin

$$\vec{D}_{1n} = (\vec{D}_1 \cdot \vec{a}_n) \vec{a}_n$$

$$= (1.6 - 2.4) \vec{a}_n$$

$$= (-0.64, -0.48)$$

$$\vec{D}_{1t} = \vec{D}_1 - \vec{D}_{1n}$$

$$= (2, -4, 6.5) - (-0.64, -0.48, 0)$$

$$= (2.64, -3.52, 6.5)$$

$$\vec{D}_{2n} = \vec{D}_{1n} = (-0.64, -0.48)$$

$$\vec{E}_{2t} = \vec{E}_{1t} \rightarrow \frac{D_{2t}}{\epsilon_2} = \frac{D_{1t}}{\epsilon_1}$$

$$\vec{D}_{2t} = \frac{\epsilon_2}{\epsilon_1} \vec{D}_{1t}$$

$$= \frac{2.5}{1} (2.64, -3.52, 6.5)$$

$$= (6.6, -8.8, 16.25)$$

$$D_2 = D_{2n} + D_{2t}$$

$$= (5.96, -9.28, 16.25) \text{ nC/m}^2$$

$$\theta_2 = \cos^{-1} \left( \frac{D_{2n} \cdot \vec{a}_n}{|D_2|} \right)$$

$$= 87.66^\circ$$

41) . reg 1:  $\epsilon_1 = 2\epsilon_0$ ,  $x + 2y + z = 1$  <sup>condition</sup>  
 . reg 2:  $\epsilon_2 = 5\epsilon_0$   
 $\vec{E}_1 = (20, -10, 40) \text{ V/m}$   
 a) normal & tangential components of  $\vec{E}_1$  &  $\vec{E}_2$  b)

a)  $f(x, y, z) = x + 2y + z - 1 = 0$

$$\nabla f = (1, 2, 1)$$

$$\vec{a}_n = \frac{\nabla f}{|\nabla f|} = \frac{1}{\sqrt{6}} (1, 2, 1)$$

$$\begin{aligned} \vec{E}_{in} &= (\vec{E}_1 \cdot \vec{a}_n) \vec{a}_n \\ &= \frac{1}{\sqrt{6}} (20 - 20 + 40) \frac{1}{\sqrt{6}} (1, 2, 1) \\ &= (6.67, 13.3, 6.67) \text{ V/m} \end{aligned}$$

$$\begin{aligned} \vec{E}_{it} &= \vec{E}_1 - \vec{E}_{in} \\ &= (13.3, 23.3, 33.3) \text{ V/m} \end{aligned}$$

b)  $\vec{E}_{2t} = \vec{E}_{1t} = (13.3, -23.3, 33.3)$

$$\vec{D}_{2n} = \vec{D}_{1n}$$

$$\epsilon_2 \vec{E}_{2n} = \epsilon_1 \vec{E}_{1n}$$

$$\vec{E}_{2n} = \frac{\epsilon_1}{\epsilon_2} \vec{E}_{1n}$$

$$= \frac{2}{5} (6.67, 13.3, 6.67)$$

$$= (2.7, 5.3, 3.3)$$

$$\vec{E}_2 = \vec{E}_{2t} + \vec{E}_{2n}$$

$$= (16, -18, 36) \text{ V/m}$$

①  $\uparrow \epsilon_1 = 3\epsilon_0$

②  $\epsilon_2 = 4.5\epsilon_0$

42)  $\vec{E}_1 = (10, -6, 12) \text{ V/m}$  in fig 5.22  $\rightarrow$   
 a)  $\vec{P}_1 = ?$ , b)  $\vec{E}_2$  &  $\theta_{\text{axis}}$  of  $\vec{E}_2 = ?$  c) energy density of regions

a)  $\vec{P}_1 = \chi_{e1} \epsilon_0 \vec{E}_1$

$$= 2 \epsilon_0 (10, -6, 12)$$

$$= (6.177, 0.106, 0.212) \text{ nC/m}^2$$

b)  $\vec{E}_{1n} = -6 \vec{a}_y$

$$\vec{E}_{2t} = \vec{E}_{1t} = 10 \vec{a}_x + 12 \vec{a}_z$$

$$\vec{D}_{2n} = \vec{D}_{1n}$$

$$\epsilon_2 \vec{E}_{2n} = \epsilon_1 \vec{E}_{1n}$$

$$\vec{E}_{2n} = \frac{\epsilon_1}{\epsilon_2} \vec{E}_{1n}$$

$$= \frac{3}{4.5} (-6 \vec{a}_y)$$

$$= -4 \vec{a}_y$$

$$\vec{E}_2 = \vec{E}_{2t} + \vec{E}_{2n}$$

$$= (10, -4, 12) \text{ V/m}$$

$$\tan \theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{10^2 + 12^2}}{4} = (3.905) \tan^{-1} \rightarrow \theta_2 = 75.64^\circ$$

c)  $W_E = \frac{1}{2} \vec{D} \cdot \vec{E}$

$$= \frac{1}{2} \epsilon \vec{E} \cdot \vec{E}$$

$$= \frac{1}{2} \epsilon (\vec{E})^2$$

$$W_{E1} = \frac{1}{2} \epsilon_1 (\vec{E}_1)^2$$

$$= \frac{1}{2} (3\epsilon_0) (10^2 + 6^2 + 12^2)$$

$$= 3.71 \text{ nJ/m}^3$$

$$W_{E2} = \frac{1}{2} \epsilon_2 (\vec{E}_2)^2$$

$$= \frac{1}{2} (4.5\epsilon_0) (10^2 + 4^2 + 12^2)$$

$$= 5.17 \text{ nJ/m}^3$$

## Chapter 6

2)  $V = \frac{1}{r^2} 10 \cos \theta \sin \phi$   $\epsilon = \epsilon_0$  a)  $\vec{E}$  @  $P(1, 60^\circ, 30^\circ)$  b)  $\rho_v$  @  $P$

a)  $-\vec{E} = \nabla V = \left( \frac{\partial}{\partial r} \left( \frac{10 \cos \theta \sin \phi}{r^2} \right), \frac{1}{r^2} \frac{\partial}{\partial \theta} (10 \cos \theta \sin \phi), \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \phi} (10 \cos \theta \sin \phi) \right)$   
 $-\vec{E} @ P = \left( \frac{\partial}{\partial r} \left( \frac{10 \cos \frac{\pi}{3} \sin \frac{\pi}{6}}{r^2} \right), \frac{1}{r^2} \frac{\partial}{\partial \theta} (10 \cos \theta \sin \frac{\pi}{6}), \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \phi} (10 \cos \theta \sin \frac{\pi}{6}) \right)$   
 $= \left( 5, -4.33, -5 \right) \text{ V/m}$

$$\frac{1}{\sin \theta} (\sin \theta)^2 = 2 \cdot (\cos \theta)^1$$

b)  $-\frac{\rho_v}{\epsilon} = \nabla^2 V$

$$\begin{aligned} \rho_v &= -\epsilon \nabla^2 V \\ &= -\epsilon \left( \frac{1}{r^2} \frac{\partial}{\partial r} \left( \frac{\partial}{\partial r} \left( \frac{10 \cos \theta \sin \phi}{r^2} \right) \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin^2 \theta \frac{\partial}{\partial \theta} \left( \frac{10 \cos \theta \sin \phi}{r^2} \right) \right) - \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \left( \frac{10 \cos \theta \sin \phi}{r^2} \right) \right) \\ &= -\epsilon \left( \frac{1}{r^2} \frac{\partial}{\partial r} \left( \frac{-20 \cos \theta \sin \phi}{r^3} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin^2 \theta \frac{-10 \sin \theta}{r^2} \right) - \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} (10 \cos \theta \sin \phi) \right) \\ &= -\epsilon \left( \frac{1}{r^4} 20 \cos \theta \sin \phi - \frac{1}{r^3 \sin \theta} 20 \cos \theta \sin \phi - \frac{1}{r^4 \sin^2 \theta} 10 \cos \theta \sin \phi \right) \\ &= -\epsilon \left( \frac{-10 \cos \theta \sin \phi}{r^4 \sin^2 \theta} \right) \\ &= +\epsilon_0 10 \cos \theta \sin \phi \\ &\quad \frac{1}{r^4 \sin^2 \theta} \end{aligned}$$

@  $P$   $= \frac{+ \epsilon_0 10 \cos \frac{\pi}{3} \sin \frac{\pi}{6}}{1^4 \sin^2 \frac{\pi}{3}}$   
 $= 29.5 \text{ pC/m}^3$



15) cylindrical struct in fig. 6.31 w/  $r_i = 5 \text{ mm}$ ,  $r_o = 15 \text{ mm}$ .  $V(p=5 \text{ mm}) = 100 \text{ V}$  &  $V(p=15 \text{ mm}) = 0 \text{ V}$ .  
 $V = ?$ ,  $\vec{E} = ?$ ,  $\vec{D} = ?$  @  $p = 10 \text{ mm}$  &  $p_s$  on each plate.  $\epsilon_r = 2$

$$V = \frac{V_o \ln(p/a)}{\ln(b/a)} = V_o \frac{\ln(\frac{a}{p})}{\ln(\frac{a}{b})}$$

$$\vec{E} = -\nabla V = \frac{-V_o}{p \ln(b/a)} \vec{a}_p = \frac{+V_o}{p \ln(a/b)} \vec{a}_p$$

$$\vec{D} = \epsilon \vec{E} = -\frac{\epsilon_0 \epsilon_r V_o}{p \ln(b/a)} \vec{a}_p$$

$$\rho_s = D_n = \pm \frac{\epsilon_0 \epsilon_r V_o}{p \ln(b/a)} \Big|_{p=a,b}, \quad V_o = 100 \text{ V}, \quad b = 5 \text{ mm}, \quad a = 15 \text{ mm}, \quad \epsilon_r = 2, \quad p = 10 \text{ mm}$$

$$V = \frac{100 \ln(10/15)}{\ln(5/15)} = 36.91 \text{ V}$$

$$\vec{E} = \frac{100}{10 \text{ mm} \ln 3} \vec{a}_p = 9.1 \vec{a}_p \text{ kV/m}$$

$$\vec{D} = 9.1 \text{ k} \cdot \epsilon_0 \cdot 2 \vec{a}_p = 161 \vec{a}_p \text{ nC/m}^2$$

$$\rho_s(p=r_i=5 \text{ mm}) = \epsilon_0 \cdot 2 \cdot \frac{10^5}{5 \ln 3} = 332 \text{ nC/m}^2$$

$$\rho_s(p=r_o=15 \text{ mm}) = \epsilon_0 \cdot 2 \cdot \frac{10^5}{15 \ln 3} = -107.3 \text{ nC/m}^2$$

16) in cylindrical coords,  $V = 50V$  on plane  $\phi = \pi/2$  &  $V = 0$  on plane  $\phi = 0$ . planes along  $z$ -axis.  $\vec{E}$  on planes =?

$$0 = \frac{1}{\rho} \frac{d^2 V}{d\phi^2}$$

$$0 = \frac{d^2 V}{d\phi^2}$$

$$A = \frac{dV}{d\phi}$$

$$V = A\phi + B$$

$$0 = A(0) + B \rightarrow B = 0$$

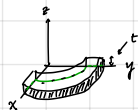
$$50 = A\left(\frac{\pi}{2}\right) + 0 \rightarrow A = \frac{100}{\pi}$$

$$\vec{E} = -\nabla V$$

$$= -\frac{1}{\rho} \frac{dV}{d\phi} \hat{a}_\phi$$

$$= -\frac{1}{\rho} A \hat{a}_\phi$$

$$\vec{E} = -\frac{1}{\rho} \cdot \frac{100}{\pi} \hat{a}_\phi$$



29) Show that resistance of bar of fig 6.17 between the vertical ends located at  $\phi = 0$  &  $\phi = \frac{\pi}{2}$  is  $R = \frac{\pi}{2\sigma t \ln \frac{b}{a}}$   
 • if corners @  $\phi = 0, \frac{\pi}{2}$  maintained @ potent. diff. of  $V_0$ .

$$E_\phi = \frac{2V_0}{\pi\rho}, \quad J = \sigma E$$

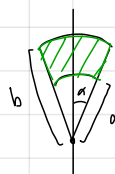
$$I = \int J \cdot dS$$

$$= \frac{2V_0 \sigma}{\pi} \int_{\rho=a}^b \int_{z=0}^t \frac{1}{\rho} \rho d\rho dz$$

$$= \frac{2V_0 \sigma}{\pi} \cdot t \cdot \ln(b/a)$$

$$R = V_0 / I$$

$$= \frac{\pi}{2\sigma t \ln(b/a)}$$



30) Show  $R$  of spherical shell sector of  $\sigma$  w/ cross section in fig. 31 where  $(0 \leq \phi \leq 2\pi)$  between its base (from  $r = a \rightarrow b$ ) is

$$R = \frac{1}{2\pi\sigma(1-\cos\alpha)} \left( \frac{1}{a} - \frac{1}{b} \right)$$

$$\text{if } V(r=a) = 0, \quad V(r=b) = V_0$$

$$E = \frac{V_0}{r^2(1/a - 1/b)}, \quad J = \sigma E$$

$$I = \int J \cdot dS = \frac{V_0 \sigma}{1/a - 1/b} \int_{\theta=0}^{\alpha} \int_{\phi=0}^{2\pi} \frac{1}{r^2} r^2 \sin\theta d\theta d\phi$$

$$= \frac{V_0 \sigma}{1/a - 1/b} \cdot 2\pi \cdot (-\cos\theta) \Big|_0^{\alpha}$$

$$R = \frac{V_0}{I}$$

$$= \frac{\frac{1}{a} - \frac{1}{b}}{2\pi\sigma(1-\cos\alpha)}$$

31) spherical shell  $r_i = a$  &  $r_o = b$ . uniform  $\sigma$  & copper electrodes plated on inner & outer surfaces. Show that:

$$R = \frac{1}{4\pi\sigma} \left( \frac{1}{a} - \frac{1}{b} \right)$$

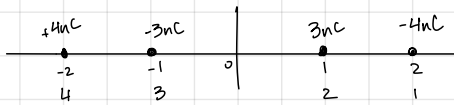
same as 6.30 but  $\alpha = \pi$

$$R = \frac{\frac{1}{a} - \frac{1}{b}}{2\pi\sigma(1 - \cos\pi)} = \frac{\frac{1}{a} - \frac{1}{b}}{2\pi\sigma(1 - (-1))} = \frac{\frac{1}{a} - \frac{1}{b}}{4\pi\sigma}$$

66) 2 pt charges:  $3\text{nC}$  @  $(0,0,1\text{m})$  &  $-4\text{nC}$  @  $(0,0,2\text{m})$  while an  $\infty$  conducting plane @  $z=0$ .

a) total charge induced on the plane

b) magnitude of  $\vec{F}_{\text{attraction}}$  between the charges & the plane



a)  $Q_i = -(3\text{nC} - 4\text{nC}) = +1\text{nC}$

b)  $\vec{F} = \vec{F}_{13} + \vec{F}_{14} + \vec{F}_{23} + \vec{F}_{24}$

$$|\vec{F}| = \frac{Q_i^2}{4\pi\epsilon_0} \left[ \frac{3 \cdot 3}{2^2} + \frac{4 \cdot 4}{4^2} - 2 \cdot \frac{3 \cdot 4}{3} \right]$$

$$= \frac{1\text{n}^2}{4\pi\epsilon_0} \left[ \frac{9}{4} + \frac{16}{16} - \frac{24}{3} \right]$$

$$= 5.25 \text{ nN}$$

## Chapter 7

1) a) biots savarts law

b) The y & z axes carry filamentary currents 10A along  $\vec{a}_y$  & 20A along  $-\vec{a}_z$   
 $\vec{H}$  @  $(-3, 4, 5)$

b)  $\vec{H} = \vec{H}_y + \vec{H}_z$

$$\vec{H}_z = \frac{I}{2\pi\rho} \vec{a}_\phi, \quad \rho = \sqrt{(-3)^2 + 4^2} = 5, \quad \vec{a}_\phi = -\vec{a}_z \times \frac{-3\vec{a}_x + 4\vec{a}_y}{5} = \frac{3\vec{a}_y - 4\vec{a}_x}{5}$$

$$= \frac{20}{2\pi(5)} (4\vec{a}_x + 3\vec{a}_y)$$

$$= 0.509\vec{a}_x + 0.382\vec{a}_y$$

$$\vec{H}_y = \frac{I}{2\pi\rho^2} \vec{a}_\phi, \quad \rho = \sqrt{(-3)^2 + 5^2} = \sqrt{34}, \quad \vec{a}_\phi = \vec{a}_y \times \frac{-3\vec{a}_x + 5\vec{a}_z}{\sqrt{34}} = \frac{3\vec{a}_z + 5\vec{a}_x}{\sqrt{34}}$$

$$= \frac{10}{2\pi\sqrt{34}^2} (5\vec{a}_x + 3\vec{a}_z)$$

$$\vec{H} = \vec{H}_y + \vec{H}_z$$

$$= 0.74\vec{a}_x + 0.38\vec{a}_y + 0.14\vec{a}_z \quad \text{A/m}$$

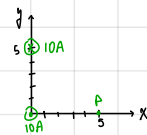
2) Straight wire w/ 2A. mag field strength = 10 mA/m. distance from wire = ?

$$H = \frac{I}{2\pi\rho}$$

$$\rho = \frac{I}{2\pi H}$$

$$= \frac{2}{2\pi \cdot 10\text{mA}}$$

$$= 31.83\text{m}$$



3) 2  $\infty$  long wires parallel to z axis, carry 10A in opposite directions like Fig 7.28.  $\vec{H} = ?$  @  $(3, 1, -2)$

$$\vec{H} = \vec{H}_1 + \vec{H}_2, \quad \vec{H}_{1,2} \text{ due to lines @ } (0, 0) \text{ & } (0, 5)$$

$$\vec{H}_1 = \frac{I}{2\pi\rho} \vec{a}_\phi, \quad \rho = 5, \quad \vec{a}_\phi = \vec{a}_z \times \vec{a}_\rho = \vec{a}_z \times \vec{a}_x = \vec{a}_y$$

$$= \frac{10}{2\pi(5)} \vec{a}_y$$

$$= \frac{1}{\pi} \vec{a}_y$$

$$\vec{H}_2 = \frac{I}{2\pi\rho} \vec{a}_\phi, \quad \rho = 5\sqrt{2}, \quad \vec{a}_\phi = \vec{a}_z \times \vec{a}_\rho, \quad \vec{a}_\rho = \frac{5\vec{a}_x - 5\vec{a}_y}{5\sqrt{2}} = \frac{\vec{a}_x - \vec{a}_y}{\sqrt{2}}$$

$$\vec{a}_\phi = -\vec{a}_z \times \frac{\vec{a}_x - \vec{a}_y}{\sqrt{2}} = \frac{-\vec{a}_y - \vec{a}_x}{\sqrt{2}}$$

$$= \frac{10}{2\pi 5\sqrt{2}} \frac{-\vec{a}_x - \vec{a}_y}{\sqrt{2}}$$

$$= \frac{1}{2\pi} (-\vec{a}_x - \vec{a}_y)$$

$$\vec{H} = \vec{H}_1 + \vec{H}_2$$

$$= \frac{1}{\pi} \vec{a}_y + \frac{1}{2\pi} (-\vec{a}_x - \vec{a}_y)$$

$$= \frac{1}{2\pi} \vec{a}_x + \frac{1}{2\pi} \vec{a}_y$$

## Chapter 7

° # 1, 2, 3, 4, 6, 7,

° # 10, 16, 17

° # 34, 37, 40, 41

4) 2 current elements:  $I_1 dl_1 = 4 \times 10^{-5} \vec{a}_x$  A.m @ (0,0,0)  
 $I_2 dl_2 = 6 \times 10^{-5} \vec{a}_y$  A.m @ (0,0,1)

$\vec{H}$  @ (3,1,-2)

$$\vec{H} = d\vec{H} = \frac{I \cdot d\vec{l} \times \vec{R}}{4\pi R^3} = \vec{H}_1 + \vec{H}_2$$

$$\vec{H}_1: \vec{R} - \vec{O} = \vec{R} \rightarrow |\vec{R}| = \sqrt{3^2 + 1^2 + 2^2} = \sqrt{14}$$

$$I d\vec{l} \times \vec{R} = 4 \times 10^{-5} \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ 1 & 0 & 0 \\ 3 & 1 & -2 \end{vmatrix} = 4 \times 10^{-5} (0, 2, 1)$$

$$\vec{H}_1 = \frac{4 \times 10^{-5} (0, 2, 1)}{4\pi (14)^{3/2}} = (0, 0.012, 0.0061) \times 10^{-5}$$

$$\vec{H}_2: \vec{R} - (0,0,1) = (3,1,-2) - (0,0,1) = (3,1,-3) \rightarrow |\vec{R}| = \sqrt{3^2 + 1^2 + 3^2} = \sqrt{19}$$

$$I d\vec{l} \times \vec{R} = 6 \times 10^{-5} \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ 0 & 1 & 0 \\ 3 & 1 & -3 \end{vmatrix} = 6 \times 10^{-5} (-3, 0, -3)$$

$$\vec{H}_2 = \frac{6 \times 10^{-5} (-3, 0, -3)}{4\pi (19)^{3/2}} = (-0.017, 0, -0.017) \times 10^{-5}$$

$$\vec{H} = \vec{H}_1 + \vec{H}_2$$

$$= [(0, 0.012, 0.0061) + (-0.017, 0, -0.017)] \times 10^{-5}$$

$$= (-0.17, 1.2, -0.11) \mu A/m$$

5) conducting filament  $\rightarrow I$  from A(0,0,a) to B(0,0,b). Show that at P(x,y,0)

$$\vec{H} = \frac{I}{4\pi \sqrt{x^2 + y^2}} \left[ \frac{b}{\sqrt{x^2 + y^2 + b^2}} - \frac{a}{\sqrt{x^2 + y^2 + a^2}} \right] \vec{a}_\phi$$

$$\vec{H} = \vec{H}_y + \vec{H}_z$$

$$\vec{H}_z = \frac{I}{2\pi \rho} \vec{a}_\phi, I = 20, \vec{p} = (-3, 4), |\vec{p}| = \sqrt{3^2 + 4^2} = 5, \vec{a}_\phi = \vec{a}_z \times \vec{a}_p = -\vec{a}_z \times \frac{-3\vec{a}_x + 4\vec{a}_y}{5} = \begin{vmatrix} 0 & 0 & -1 \\ -0.6 & 0.8 & 0 \end{vmatrix} = 0.8\vec{a}_x + 0.6\vec{a}_y$$

$$= \frac{20}{2\pi(5)} (0.8, 0.6, 0)$$

$$= (0.509, 0.382, 0)$$

$$\vec{H}_y = \frac{I}{2\pi \rho} \vec{a}_\phi, I = 10, \vec{p} = -3\vec{a}_x + 5\vec{a}_z, |\vec{p}| = \sqrt{3^2 + 5^2} = \sqrt{34}, \vec{a}_\phi = \vec{a}_z \times \vec{a}_p = \frac{1}{\sqrt{34}} \begin{vmatrix} 0 & 1 & 0 \\ -3 & 0 & 5 \end{vmatrix} = \frac{1}{\sqrt{34}} (5, 0, 3)$$

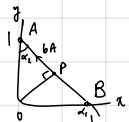
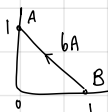
$$= \frac{10}{2\pi(34)} (5, 0, 3)$$

$$= (0.23, 0, 0.14)$$

$$\vec{H} = \vec{H}_y + \vec{H}_z$$

$$= (0.74, 0.382, 0.14) A/m$$

6) AB in Fig 7.29 in dec circuit.  $\vec{H}$  @ origin due to AB.



$$\vec{H} = \frac{I}{4\pi \rho} (\cos \alpha_2 - \cos \alpha_1) \vec{a}_\phi, \alpha_1 = 135^\circ, \alpha_2 = 45^\circ, \rho = \frac{1}{2}\sqrt{2} = \frac{\sqrt{2}}{2}, \vec{a}_\phi = \vec{a}_z \times \vec{a}_p = \frac{-a_x + a_y}{\sqrt{2}} \times \frac{-a_x - a_y}{\sqrt{2}} = \frac{1}{2} \begin{vmatrix} -1 & 1 & 0 \\ -1 & -1 & 0 \end{vmatrix} = \vec{a}_z$$

$$= \frac{b}{4\pi \frac{\sqrt{2}}{2}} (\cos 45^\circ - \cos 135^\circ) \vec{a}_z$$

$$= \frac{3}{\pi} \vec{a}_z$$

$$\vec{H}(0,0,0) = 0.95 A/m$$

7) line  $x=0, y=0, 0 \leq z \leq 10\text{m}$  carries  $2\text{A}$  along  $\hat{a}_z$ .  $\vec{H} = ?$  @ points...

a)  $(5, 0, 0)$

b)  $(5, 5, 0)$

c)  $(5, 15, 0)$

d)  $(5, -15, 0)$

a)  $\rho = 5, \hat{a}_\rho = \hat{a}_y, \cos \alpha_1 = 0, \cos \alpha_2 = \frac{10}{\sqrt{25}}$ .

$$\vec{H} = \frac{2}{4\pi(5)} \frac{10}{\sqrt{25}} \hat{a}_y = 28.41 \hat{a}_y \text{ mA/m}$$

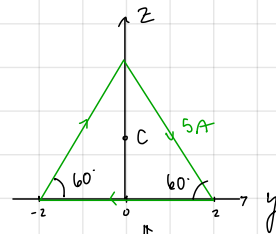
b)  $\rho = 5\sqrt{2}, \hat{a}_\rho = \hat{a}_z \times \frac{\hat{a}_x + \hat{a}_y}{\sqrt{2}} = \frac{-\hat{a}_x + \hat{a}_y}{\sqrt{2}}, \cos \alpha_1 = 0, \cos \alpha_2 = \frac{10}{\sqrt{50}}$

$$\vec{H} = \frac{2}{4\pi 5\sqrt{2}} \frac{10}{\sqrt{50}} \frac{-\hat{a}_x + \hat{a}_y}{\sqrt{2}} = 13(-\hat{a}_x + \hat{a}_y) \text{ mA/m}$$

c)  $\rho = \sqrt{250} = 5\sqrt{10}, \cos \alpha_1 = 0, \cos \alpha_2 = \frac{10}{\sqrt{550}}, \hat{a}_\rho = \hat{a}_z \times \frac{5\hat{a}_x + 15\hat{a}_y}{5\sqrt{10}} = \frac{5\hat{a}_y - 15\hat{a}_x}{5\sqrt{10}}$

$$\vec{H} = \frac{2}{4\pi 5\sqrt{10}} \cdot \frac{10}{\sqrt{550}} \cdot \frac{-15\hat{a}_x + 5\hat{a}_y}{\sqrt{10}} = -5.1\hat{a}_x + 1.7\hat{a}_y \text{ mA/m}$$

d) by symmetry:  $\vec{H} = 5.1\hat{a}_x + 1.7\hat{a}_y \text{ mA/m}$



10)  $\vec{H} = ?$  @ center of C of an equilateral triangular loop of side  $4\text{m}$  carrying  $5\text{A}$  like in fig. 7.31

for side on  $y$  axis,

$$\vec{H}_1 = \frac{I}{4\pi\rho} (\cos \alpha_2 - \cos \alpha_1) \hat{a}_\phi, \hat{a}_\phi = -\hat{a}_x, \rho = 2 \tan 30^\circ \cdot \frac{2}{\sqrt{3}}, \alpha_2 = 30^\circ, \alpha_1 = 150^\circ$$

$$= \frac{5}{4\pi} \frac{\sqrt{3}}{2} (\cos 30^\circ - \cos 150^\circ) (-\hat{a}_x) = \frac{-15}{8\pi} \hat{a}_x$$

$$\vec{H} = 3\vec{H}_1 = -1.79 \hat{a}_x \text{ A/m}$$

11b) 2 id. loops parallel by dist.  $d$  (fig 7.35).  $\vec{H} = ?$  @  $(0, 0, d)$ .  $a=3\text{cm}, d=4\text{cm}, I=10\text{A}$ .

$$\vec{H}_1 = \frac{I}{2a} \hat{a}_z, \vec{H}_2 = \frac{I a^2}{2(a^2 + d^2)^{3/2}} \hat{a}_z$$

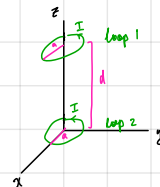
$$\vec{H} = \vec{H}_1 + \vec{H}_2$$

$$= \frac{I}{2} \left[ \frac{1}{a} + \frac{a^2}{(a^2 + d^2)^{3/2}} \right] \hat{a}_z$$

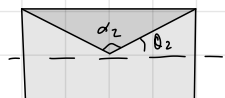
$$= \frac{10}{2} \left[ \frac{1}{3 \cdot 10^{-2}} + \frac{3 \cdot 10^{-2}}{(3^2 + 4^2)^{3/2} \cdot 10^{-2}} \right] \hat{a}_z$$

$$= 500 \left( \frac{1}{3} + \frac{9}{125} \right) \hat{a}_z$$

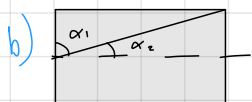
$$= 202.7 \text{ A/m}$$



17) solenoid of  $r=4\text{mm}$  &  $l=2\text{m}$  w/ 150 turns/m w/ 500mA. a)  $|\vec{H}|$  at center. b)  $|\vec{H}|$  at ends of solenoid



$$\begin{aligned} \text{a) } |\vec{H}| &= \frac{nI}{2} (\cos\theta_2 - \cos\theta_1), \quad \cos\theta_2 = -\cos\theta_1 = \frac{l/2}{\sqrt{a^2 + l^2/4}} \\ &= \frac{nI l}{2\sqrt{a^2 + l^2/4}} = \frac{0.5 \times 150 \times 2 \times 10^{-2}}{2 \times 10^{-3} \times \sqrt{4^2 + 10^2}} = 69.6 \text{ A/m} \end{aligned}$$



$$\alpha_1 = 90^\circ, \quad \tan\theta_2 = \frac{a}{b} = \frac{4}{20} = 0.2 \rightarrow \theta_2 = 11.31^\circ$$

$$|\vec{H}| = \frac{nI}{2} \cos\theta_2 = \frac{150 \times 0.5}{2} \cos 11.31^\circ = 36.77 \text{ A/m}$$

34) free space:  $\vec{B} = y^2 \vec{a}_x + z^2 \vec{a}_y + x^2 \vec{a}_z$  Wb/m<sup>2</sup>

a) Show  $\vec{B}$  is mag field

b) mag flux:  $x=1, 0 < y < 1, 1 < z < 4$

c)  $\vec{J} = ?$

d) total mag flux thru cube surface of  $0 < x, y, z < 2$

$$\begin{aligned} \text{a) } 0 &= \nabla \cdot \vec{B} \\ &= \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \end{aligned}$$

$$\begin{aligned} \text{b) } dS &= dy dz \vec{a}_x \\ \Psi &= \int \vec{B} \cdot d\vec{S} \\ &= \int_{z=1}^4 \int_{y=0}^1 y^2 dy dz \\ &= \frac{1}{3} y^3 \Big|_0^1 \cdot z \Big|_1^4 \\ &= 1 \text{ Wb} \end{aligned}$$

$$\begin{aligned} \text{c) } \vec{J} &= \nabla \times \vec{H} = \nabla \times \frac{\vec{B}}{\mu_0} \\ &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & z^2 & x^2 \end{vmatrix} \cdot \frac{1}{\mu_0} \\ &= (-2z\vec{a}_x - 2x\vec{a}_y - 2y\vec{a}_z) / \mu_0 \\ &= \frac{-2}{\mu_0} (z, x, y) \text{ A/m}^2 \end{aligned}$$

$$\begin{aligned} \text{d) } \therefore 0 &= \nabla \cdot \vec{B} \\ \therefore \Psi &= \int_V \vec{B} \cdot d\vec{S} = \int_V \nabla \cdot \vec{B} dV = 0 \end{aligned}$$

37) free space,  $\vec{B} = \frac{1}{\rho} 20 \sin^2 \phi \vec{a}_z$  Wb/m<sup>2</sup>. mag flux thru strip:  $z=0$ ,  $1 < \rho < 2$ ,  $0 < \phi < \pi/4$

$$\begin{aligned}\Psi &= \int_S \vec{B} \cdot d\vec{S} \\ &= \int_{\phi=0}^{\pi/4} \int_{\rho=1}^2 \frac{1}{\rho} 20 \sin^2 \phi \cdot \rho \cdot d\rho d\phi \\ &= 20 \cdot \int_0^{\pi/4} \sin^2 \phi \int_1^2 \frac{1}{\rho} d\rho \\ &= 20 \cdot \int_0^{\pi/4} \frac{1}{2} (1 - \cos(2\phi)) \\ &= 20 \cdot \frac{1}{2} \cdot \left( \phi - \frac{1}{2} \sin(2\phi) \right) \Big|_0^{\pi/4} \\ &= 10 \left( \frac{\pi}{4} - \frac{1}{2} \right) \\ &= 2.86 \text{ Wb}\end{aligned}$$

40 & 41) given fields, which can be electrostatic or magnetostatic field in free space

a)  $\vec{A} = y \cos ax \vec{a}_x + (y + e^{-x}) \vec{a}_z$   
 $\nabla \cdot \vec{A} \neq 0$   
 $\nabla \times \vec{A} \neq 0$  } neither elec/mag

b)  $\vec{B} = \frac{1}{\rho} 20 \vec{a}_\phi$   
 $\nabla \cdot \vec{B} = 0$   
 $\nabla \times \vec{B} = 0$  } can be E-field in charge free region

c)  $\vec{C} = r^2 \sin \theta \vec{a}_\phi$   
 $\nabla \cdot \vec{C} = 0$   
 $\nabla \times \vec{C} \neq 0$  } possibly H field

a)  $\vec{D} = y^2 z \vec{a}_x + z(x+1)y \vec{a}_y - (x+1)z^2 \vec{a}_z$   
 $\nabla \cdot \vec{D} = 0$   
 $\nabla \times \vec{D} \neq 0$  } possibly mag field

b)  $\vec{E} = \frac{1}{\rho} (z+1) \cos \phi \vec{a}_\rho + \frac{1}{\rho} \sin \phi \vec{a}_z$   
 $\nabla \cdot \vec{E} = 0$   
 $\nabla \times \vec{E} \neq 0$  } possibly mag field

c)  $\vec{F} = \frac{1}{r^2} (2 \cos \theta \vec{a}_r + \sin \theta \vec{a}_\theta)$   
 $\nabla \cdot \vec{F} \neq 0$   
 $\nabla \times \vec{F} \neq 0$  } neither elec/mag field

## Chapter 8 Questions

### ◦ Chapter 8

- # 1, 2, 3, 4, 5, 6, 7, 8, 9
- # 18, 19, 20, 21, 24, 25, 26
- # 30, 31, 32, 33, 34
- # 42, 43, 44, 45, 46, 50, 51